

Automated Vehicles- Framing, Transparency and Liability (#126135)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has 2 authors.

Pre-registered on:

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1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

In a previous study, we found that framing autonomous vehicles (AVs) as 'Autopilot' leads to higher perceived capability and liability of an AV firm for accidents that occur while the software is engaged, as compared to framing the software as 'Copilot'.

In this study, we test how participants' judgments are affected by whether we reveal that the true capability of the Autopilot and Copilot systems are exactly the same--Level 2 autonomy. The main hypothesis of this study is that, while transparency about the system's objective capability of AVs can reduce the inflated perceptions of responsibility for a system labeled as autopilot vs. copilot, people will still hold the company more responsible and liable for damages if it labels its system as Autopilot as opposed to Copilot.

3) Describe the key dependent variable(s) specifying how they will be measured.

The dependent variables are (1) perceived level of automation, (2) the level of responsibility assigned to the driver or the AV software in case of an accident and (3) the level of liability the firm or driver is held to in case of an accident. We will measure preferred levels of autonomous vehicles on a 6-point scale with endpoints, 1 - Level 1 automation (not automated at all) and 6 - Level 6 automation (fully automated). We will measure the level of responsibility with 2 questions on a 100-point scale with endpoints, 0 - completely disagree that the AV system (human driver) is responsible for the accident and 100 - completely agree that the AV system (human driver) is responsible for the accident. We will measure the level of liability with 2 questions on a 100-point scale with endpoints, 0 - completely disagree that the company (human driver) is liable for damages in the accident and 100 - completely agree that the company (human driver) is liable for damages in the accident.

At the end of the survey, we will include an individual differences scale on perceptions of misleading and deceptive marketing practices for products and promotions.

4) How many and which conditions will participants be assigned to?

This will be a 2 (label: autopilot, copilot) x 2 (transparency: present, absent) design. Participants will be randomly assigned to answer questions about the AV labeled either Autopilot or Copilot and they will either receive information on the actual level of automation for the AV or not, which is the transparency condition. Therefore, the possible conditions are- Autopilot & full transparency; Copilot & full transparency; Autopilot & zero transparency; Copilot & zero transparency.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

We will conduct factorial ANOVA tests with the scores on the responsibility and liability scales as the outcome variables and the label and level of transparency as the predictors. For both tests, we predict main effects of label and level of transparency. There will also be an interaction effect test to investigate the effect of the interaction between the label and the transparency condition on the outcome variables, but we are agnostic to these results.

Regardless of the ANOVA result, we will conduct t-tests comparing the DV (liability or responsibility ratings) between the autopilot and copilot conditions. We will run this t-test for both the transparency present and absent conditions, i.e., four t-tests in total.

We will run a parallel mediation analysis with perceived automation capability as the mediator, the label as the predictor variable and the responsibility and liability measures as the outcome variables.

We will also run moderated mediation models, in which the b path of the above mediation model is moderated by individual differences in perceptions of misleading/deceptive marketing practices.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

We will exclude participants who fail any 1 of our 2 attention check questions incorrectly and who fail any 1 of our 2 comprehension check questions incorrectly.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We will collect responses from 1000 participants.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

We will include some demographic questions but nothing identifiable (age, gender, whether they have a drivers license). We will also ask participants how familiar they are with AVs on a 100-point scale with endpoints, 0- Very little and 100- A lot. These will be included as covariates in additional exploratory analyses as robustness checks.